
Template RPNG Tech Report

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RPNG

Robot Perception and Navigation Group (RPNG)
Tech Report - RPNG-2018-EXAMPLE
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1 Introduction

This is an example tech report that is the template for the group. To ensure uniformity, please do not change the structure of these latex documents. There are a few examples for common things to use etc.

2 Latex Examples

2.1 Lists

- A simple list.
- With no spacing between each bullet.
- Change to “enumerate” to get a numbered list.
- Custom spacing.
- One “em” is the height of a normal text line.

2.2 Equations and Custom Commands

$${}^G\mathbf{p}_{k+1} = {}^G\mathbf{p}_k + \int_k^{k+1} \mathbf{R}_{(G\bar{q})}^{\tau L} \mathbf{v}_m d\tau \quad (1)$$

Here is some nice inline text, that doesn’t break the align! Use the “vect” and “skewsym” commands to make your equations cleaner.

$$= \boldsymbol{\alpha} * \mathbf{x} + \mathbf{c} \quad (2)$$

$$= \lfloor \mathbf{d} \times \rfloor + \mathbf{c} \quad (3)$$

2.3 Text Properties

- Smaller text size, note brackets bound the applied area.
- Larger text size, note brackets bound the applied area.
- **Make text a specific color.**
- **Make text bold.**
- *Make text italicized.*

2.4 Figures and Plots

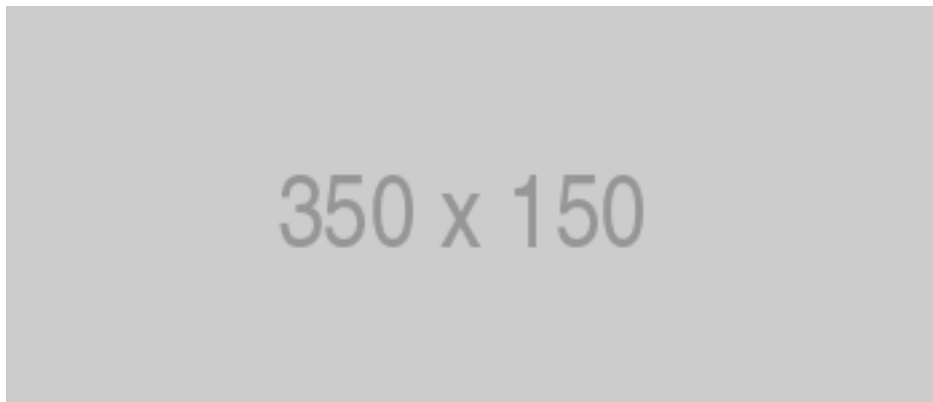


Figure 1: Simple figure with label.

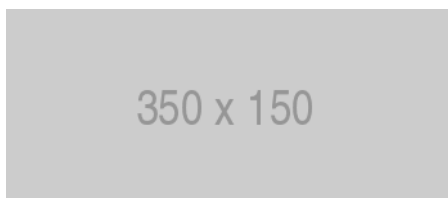


Figure 2: Simple figure with label.



Figure 3: Total figure caption.

2.5 Citations and References

Here is an example citation [1] where we can do a couple [2, 3]. We can also reference Figure 1 and equation (3). The bibliography will be automatically created, and ordered in the order things are referenced. That should be about all one would need to do.

Appendix A: Cool Appendix Title

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

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References

- [1] Kevin Eickenhoff, Patrick Geneva, and Guoquan Huang. “High-Accuracy Preintegration for Visual-Inertial Navigation”. In: *in Proc. of International Workshop on the Algorithmic Foundations of Robotics*. 2016.
- [2] G. Huang, M. Kaess, and J. Leonard. “Towards Consistent Visual-Inertial Navigation”. In: *Proc. of the IEEE International Conference on Robotics and Automation*. Hong Kong, China, 2014, pp. 4926–4933.
- [3] Kevin Eickenhoff and Guoquan Huang. “State-Transition and Observability Constrained EKF for Multi-robot Cooperative Localization”. In: *Proc. of the American Control Conference*. (submitted). Chicago, IL, 2015.